COMMERCIAL MORTGAGE-BACKED SECURITIES (CMBS)

Learning Outcome

- describe characteristics and risks of commercial mortgage-backed securities

Commercial mortgage-backed securities (CMBS) are backed by a pool of commercial mortgages on income-producing property, such as multifamily properties (e.g., apartment buildings), office buildings, industrial properties (including warehouses), shopping centers, hotels, and health care facilities (e.g., senior housing care facilities). Repayment can be made from leases and other revenue the property generates. The collateral is a pool of commercial loans that were originated either to finance a commercial property acquisition or to refinance a prior mortgage obligation. Exhibit 7 provides an example of a CMBS structure where the mortgages on several properties owned by the hypothetical car manufacturer Bright Wheel Automotive (BRWA), used in previous lessons, are securitized.

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Exhibit 7:

Sequential Pay CMBS, Summary of Terms

CMBS SPE
Commercial Finance Partners

Assets	Liabilities
WAC = 4.66% WAM = 88 mos.	GBP 100 million
Office Bldg: GBP 21m FMV: GBP 27m, 4.0% Maturity: 5 Yrs	**Tranche A: GBP 25m** 2.5%, Maturity 3 Yrs
Factory Bldg: GBP 32m FMV: GBP 40m, 6.0% Maturity: 5 Yrs	**Tranche B: GBP 25m** 4.25%, Maturity 10 Yrs
Office Bldg: GBP 30m FMV: GBP 36m, 3.5% Maturity: 10 Yrs	**Tranche C: GBP 25m** 4.75%, Maturity 10 Yrs
Warehouse: GBP 17m FMV: GBP 20m, 5.0% Maturity: 10 Yrs	**Tranche D: GBP 25m** 5.75%, Maturity 10 Yrs

▶ Expand / collapse Extended Description

In this specific transaction, properties valued at GBP123 million serve as collateral for GBP100 million mortgages, giving an LTV of 81.3%, which provides GBP23 million excess collateral for the mortgage lender. The lender then transfers the GBP100 million mortgages to an independent entity, Commercial Finance Associates, which then through the securitization process issues securities backed by the mortgages. Effectively, the weighted average proceeds from the mortgages (WAMP) are 4.66% and the weighted coupon payment 4.075%, where the spread provides compensation for arranging the securitization. For simplicity, all the mortgages are fully amortizing; in practice, commercial mortgages are balloon loans and not fully amortizing.

to make these payments, and the mortgages do not allow any prepayment during their first 5 years.

CMBS transactions are common in many markets. In the United States, CMBS transactions pool a larger number of commercial mortgages. Some transactions focus on mortgages on specific property types (such as warehouses and multi-family residential) or specific geographic areas (such as the Northeast or the Southwest United States). Others pool commercial mortgages from one lender. A small, but sizable, portion of the US market focuses on securitizing a single loan backed typically by one high value, marquee property in a major city. Akin to these single asset deals are single borrower deals where the CMBS transaction securitizes multiple commercial mortgage loans from one borrower.

Outside the United States, the volume of CMBS transactions is growing, particularly in Europe. European CMBS transactions have some structural differences reflecting differences in legal standards, commercial credit conventions, and bankruptcy. For instance, many European CMBS securitize a few loans in each transaction, but these loans can originate from different European countries. This introduces legal risk to the transaction because in the event of foreclosure or bankruptcy, the sale of the property must follow local rules that can be vastly different across Europe. Typically, European CMBS carry a floating rate, but that rate may be capped, while US CMBS usually pay a fixed rate and only occasionally offer a floating rate. Overall, CMBS deals require more direct understanding of the legal, financial, and commercial structures underlying each mortgage in the securitization transaction.

CMBS Structure

Both the CMBS and RMBS securitization processes and securitization structures, respectively are highly similar. However, two features specific to CMBS structures are worth mentioning: the presence of call protection and the balloon maturity provision.

Call Protection

A critical investment feature that distinguishes CMBS from RMBS is the protection against early prepayments. Call protection is available to investors in CMBS, which results in CMBS trading more like corporate bonds than RMBS. The call protection comes either at the structure level or at the loan level.

Structural call protection is achieved through sequential-pay tranches in the CMBS: A lower-rated tranche cannot be paid down until the higher-rated tranche is completely retired. Principal losses are always borne by the junior tranches first.

Call protection on the individual loan level relies on three mechanisms, usually through covenants. The first is prepayment lockout, a contractual agreement that prohibits any prepayments during a specified period. Second, the loan agreement can also stipulate prepayment penalty points, predetermined penalties that a borrower who wants to refinance must pay to the CMBS SPE. A point is equal to 1.00% of the outstanding loan balance.

Third, **defeasance** is a mechanism that allows prepayment, but the borrower must purchase a portfolio of government securities that fully replicates the cash flows of the remaining scheduled principal and interest payments, including the balloon loan balance, on the loan. When the last obligation is paid off, the value of the portfolio is zero (that is, no funds remain). The cost of assembling such a portfolio is the cost of defeasing the loan that must be paid by the issuer.

Balloon Maturity Provision

Another material difference lies in the structure of commercial mortgages. Unlike most residential mortgages, which are self-amortizing, commercial real estate loans are not fully amortizing. They usually pay interest payments and some principal repayments throughout the loan with the balance of the unamortized principal repaid at maturity through a large

principal repayment at the maturity of the loan. There are several reasons why a borrower may fail to make the balloon payment at maturity. First, the borrower may not be able to refinance the existing mortgage to roll the balloon loan into a new loan due to the borrower's deteriorating debt service coverage or LTV ratios. Lenders also may not be able to extend the terms of the existing loan given the economic environment. Finally, the borrower may not be able to sell the property to generate sufficient funds to pay off the outstanding principal balance.

Balloon risk is the risk that the borrower fails to make the balloon payment at maturity and is in default. In this case, the lender may extend the loan over a period known as the "workout period," and modify the original terms of the loan during the workout period. Because the life of the loan is extended by the lender during the workout period, balloon risk is a type of extension risk.

CMBS Risks

Another significant difference between RMBS and CMBS is that a RMBS pool may consist of thousands of mortgages from various parts of a country, while CMBS can consist of a few underlying commercial mortgages. With RMBS collateralized by relatively homogeneous single-family residential properties, individual defaults can be frequent but have minimal impact on security holders due to the small size of any one mortgage in the RMBS pool. A single default in a CMBS pool, on the other hand, can have a significant impact on the CMBS investors. Investors in the CMBS must explicitly consider this unique concentration risk by not only analyzing the CMBS structure but also by analyzing the individual loans and properties backing the CMBS and the owners of the commercial properties themselves.

In commercial real estate lending, the key indicators of potential credit performance are the loan-to-value ratio (LTV), which was discussed earlier, and the debt service coverage ratio (DSCR or DSC ratio), like the debt-to-income ratio. The **DSC ratio** is equal to the property's annual net operating income (NOI) divided by the debt service.

$$DSC = \frac{Net\ Operating\ Income}{Debt\ Service}$$

NOI is defined as the rental income reduced by cash operating expenses and a non-cash replacement reserve reflecting the depreciation of the property over time.

$$NOI = (Rental\ income - cash\ operating\ expenses) - replacement\ reserves.$$

NOI excludes principal and interest payments on loans, capital expenditures, depreciation, and amortization.

Note that debt service is the annual amount of interest payments and principal repayments. A DSC ratio that exceeds 1.0× indicates that the cash flows from the property are sufficient to cover the debt service while maintaining the property in its initial state of repair. The higher the DSC ratio, the more likely it is that the borrower will be able to meet debt-servicing requirements from the property's cash flows.

EXAMPLE 5

Debt Service Coverage Ratio Calculation

The first property listed in Exhibit 7 has an estimated fair market value of GBP27 million. Its net operating income is GBP7,590,000 million. With annual debt service of GBP4,717,169 its DSC ratio is 1.61×, as shown below. This DSC level indicates that the property's cash flows are more than sufficient to cover the debt service while maintaining the property in its initial state of repair.

$$DSC = \frac{GBP\ 7,590,000}{GBP\ 4,717,169} = 1.61\times.$$

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debt in foreclosure.

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Exhibit 8 compares representative aspects of CMBS and RMBS.

Exhibit 8:

CMBS vs. RMBS

	CMBS	RMBS
Underlying assets	From one to a pool of commercial mortgages on income-producing property	A pool of mortgages backed by residential properties or a pool of RMBS
Issuer	▪ Lenders, commercial banks, investment banks, or syndicates of banks	▪ A government or a quasi-government entity or by a bank, financial institution, or other private business
Rate for security	▪ Europe – Floating rate; may be capped ▪ US – typically fixed rate	▪ Either fixed or floating rate
Risk aspect: Credit Risk	▪ May be high, since the assets backing the CMBS can be concentrated in one mortgage or a small number (as compared to the number backing an RMBS)	▪ Agency RMBS – Issued and fully guaranteed by the government or a quasi-government entity ▪ Non-agency RMBS – Issued by banks, financial institutions, or other private businesses that use credit enhancements to reduce credit risk
Risk aspect: Prepayment Risk	▪ *contraction risk* – Low No prepayment risk since commercial loans either do not offer a prepayment option or make prepayment uneconomical ▪ *extension risk* - High Balloon risk – Many commercial loans backing CMBS are balloon. If balloon payment not made, the lender may extend the life of the loan, leading to extension risk	▪ *contraction risk* – High Particularly during periods of declining or persistently low interest rates, or when the value of the underlying properties increase. ▪ *extension risk* – High Particularly during periods of increasing interest rates or high interest rates, or when the value of the underlying properties stays stable or declines.
Risk aspect: Default Risk	Depends on: ▪ Security held: Pool based on few mortgages (high) or RMBS pools (lower) ▪ Concentration of pool: Pool of few mortgages (high) vs. a diversified pool (lower)	▪ Aggregate risk may be lower due to diversification from many small, uniform underlying mortgages.

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1. *Determine the correct answers to complete the following sentences:*

A critical investment feature that distinguishes CMBS from RMBS is the protection against early prepayments. Call protection is available to investors in _____, which results in the securities trading more like corporate bonds than _____.

Hide Solution

Solution:

A critical investment feature that distinguishes CMBS from RMBS is the protection against early prepayments. Call protection is available to investors in *CMBS*, which results in the securities trading more like corporate bonds than *RMBS*.

2. Using the information below for a 3,000 square foot warehouse and its loan, determine if the debt service coverage (DSC) ratio in year 5 is *most likely*:

5.0%, Ten-Year CAD450,000 Loan with Twenty Year Amortization	
Rental Rate	USD20/square foot/year
Rental Income	USD60,000
Cash Operating Expenses	
▪ Property Taxes	USD6,000
▪ Insurance	USD2,000
▪ Property Maintenance	USD1,500
Depreciation	USD10,000
Annual payment (principal & interest) – years 1–10	USD36,110

A. above 1.
B. 1.
C. below 1.

Hide Solution

Solution:

The correct answer is A. The debt service coverage ratio (DSC) is calculated as:

$$DSC = \frac{Net\ Operating\ Income}{Debt\ Service}$$

$$DSC = \frac{(Rental\ Income\ -\ Cash\ Operating\ Expenses)}{Annual\ Debt\ Service\ (principal\ +\ interest\ payments)}$$

$$DSC = \frac{(60,000\ -\ 9,500)}{36,110}$$

$$DSC = \frac{USD\ 50,500}{36,110} = 1.3985\times.$$

B and C are incorrect because the calculated DSC ratio is above 1.0×, not equal to or below 1.

3. Which of the following mechanisms is *least likely* to offer investors protection from default on the individual loan level?

A. Defeasance
B. Balloon payment structure
C. Prepayment penalty points

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Solution:

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payment at maturity and is in default, then balloon risk results. Thus, the balloon payment structure increases risk to investors, rather than protecting against it. A is incorrect because defeasance, a mechanism that requires the borrower to purchase a portfolio of government securities that replicates the cash flows of the remaining scheduled principal and interest payments remaining on the loan, protects the investor. C is incorrect because prepayment penalty points (i.e., predetermined penalties that a borrower who wants to refinance must pay to the lender) offer investors call protection on the individual loan level.

4. Relative to RMBS loan pools, CMBS loan pools exhibit less:

A. credit risk.
B. prepayment risk.
C. concentration risk.

Hide Solution

Solution:

The correct answer is B. CMBS loans exhibit less prepayment risk because commercial loans either do not offer prepayment options or make prepayment uneconomical. Conversely, RMBS loans are freely prepayable, thus prepayment risk is higher. A is incorrect because CMBS loans have higher credit risk because the assets backing the CMBS can be concentrated in one mortgage or a small number of mortgages (as compared to the large number backing an RMBS). In addition, agency RMBS are issued and fully guaranteed by the government or a quasi-government entity, and non-agency RMBS are issued by banks, financial institutions, or other private businesses, which use credit enhancements to reduce credit risk. C is incorrect because CMBS loan pools exhibit more concentration risk than RMBS loan pools. CMBS loan pools exhibit a fewer number of loans and will have some larger loans that create concentration risk.

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UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{CMBS} STRUCTURED FINANCE CALENDAR

The Bloomberg CMBS function allows you to view the security description of a mortgage-backed security.

> "CMBS <GO> displays a new issue calendar for commercial mortgage-backed security deals, so you can assess past and current supply trends. You can filter the calendar to find deals that meet your specific issuance date, region, and collateral criteria." -Bloomberg

To explore this function, type CMBS into the Bloomberg command line and press Enter or <GO>. If you select a commercial mortgage-backed security from the list and then click on a sub-instrument, you will see a security description of the mortgage-backed security.

3)		ENIV 2023-CATH	CMBS	CMBS		TBA	05/22/23	150	GBP	GB	CHCM
4)	*	MFMEG 552	CMBS	MULTI4.9		TBA	05/31/23	55	USD	US	
5)		ENIV 2023-E05	CMBS	CMBS		TBA	04/24/23	175	GBP	GB	CHCM
6)	*	VCC 2023-1R	CMBS	CMBS 8.7		TBA	04/28/23	131	USD	US	CITG
7)	*	MFMEG 548	CMBS	MULTI4.7		TBA	04/11/23	65	USD	US	
8)	*	MFMEG 549	CMBS	MULTI4.9		TBA	04/13/23	26	USD	US	
9)	*	MFMEG 550	CMBS	MULTI2.9		TBA	04/28/23	471	USD	US	
10)	*	MFMEG 551	CMBS	MULTI4.9		TBA	05/17/23	36	USD	US	
11)	*	OAPBR 2023-1	CMBS	CMBS		TBA	03/16/23	13	USD	US	MF
12)		ENIV 2022-FNCH	CMBS	CMBS		TBA	03/02/23	30	GBP	GB	CHCM
13)	*	MFMEG 537	CMBS	MULTI4.8		TBA	02/24/23	9	USD	US	
14)	*	MFMEG 538	CMBS	MULTI4.4		TBA	02/28/23	75	USD	US	
15)	*	MFMEG 539	CMBS	MULTI4.3		TBA	02/28/23	50	USD	US	
16)	*	MFMEG 540	CMBS	MULTI4.7		TBA	02/28/23	35	USD	US	
17)	*	MFMEG 541	CMBS	MULTI4.6		TBA	03/16/23	95	USD	US	
18)	*	MFMEG 542	CMBS	MULTI4.7		TBA	03/16/23	91	USD	US	
19)	*	MFMEG 543	CMBS	MULTI4.5		TBA	03/31/23	75	USD	US	

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JJ Are the calculations for question 2 in the question set correct? I thought that depreciation must be deducted from net operating income in the numerator.

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